PROJECT 3 · SATELLITE RADAR TIME SERIES + DEEP LEARNING

Paddy Land Watch

What happened on each paddy field this season, and where is new filling?

AI reads a season of Sentinel-1 radar and Sentinel-2 images to tell **paddy**, **other crop**, **idle** and **filled** land apart, and ranks new filling for officers.

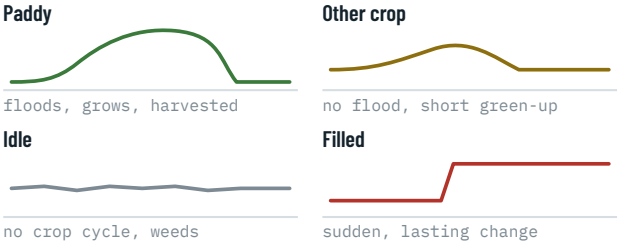
STUDY AREA	Gampaha and Kalutara, wet zone, near UoM
USERS	Agrarian Development Dept, Census and Statistics
COST	Free data and software; travel only
PLAN	Oct 2026 – Jun 2027 · 3 members

THE PROBLEM

Paddy land must by law be farmed each season, yet much lies idle or is filled, and the department finds out late.

- **The law.** Act No. 46 of 2000: paddy land "shall be cultivated with paddy during every season" (s. 30); unpermitted filling risks a Rs 100,000 fine (s. 32).
- **Idle land is large.** A February 2025 survey found over 100,000 acres of paddy fields uncultivated.
- **Filling slips through.** Kegalle audit, 2018: 372 filling cases; paddy cultivation down 76% since 1990; no legal action.
- **Money depends on it.** Yala 2026 pays Rs 25,000/ha for paddy, Rs 15,000/ha for other crops.

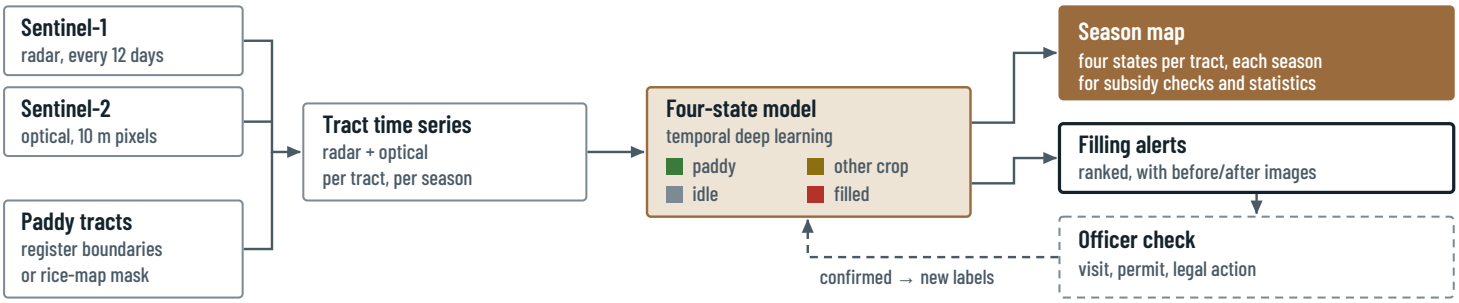
THE KEY IDEA



Schematic, one season left to right. Radar sees water and structure through cloud, optical adds greenness, and several seasons separate fallow from abandonment.

HOW IT WORKS

three inputs, a season map and an alert list out



Radar keeps the model working through monsoon cloud; each case an officer confirms becomes a new training label for the next season.

WHAT IS NEW

● covered ○ not

Rice maps exist; none we found tracks idle, other-crop and filled land each season.

STUDY	SRI LANKA	IDLE	OTHER CROP	FILLED	RADAR
Özdoğan et al. 2025 rice only	●	○	○	○	○
Sankavi et al. 2025 UoM, NCP	●	●	○	○	○
Wanninayaka et al. 2020 Ampara	●	○	○	○	●
Jayasooriya et al. 2023 Landsat	●	○	○	●	○
Kasture et al. 2025 India	○	●	○	○	●
This project	●	●	●	●	●

HOW WE PROVE IT

in held-out areas

TESTED	METRIC	AGAINST
Season map	F1 per class, macro-F1	Random Forest; AlphaEarth embeddings
Cloudy seasons	Macro-F1 when clear images are few	Optical-only model
Filling alerts	Top 20 / 50 confirmed; days to alert	Simple before/after change
District totals	% difference in sown area	Census and Statistics figures
Alert ranking	Confirmed cases per field visit	Ranking by area only

RESEARCH QUESTIONS AND OWNERS

- A — RQ1 How accurately can deep learning on Sentinel-1 and 2 time series sort each tract into paddy, other crop, idle or filled, under cloud?
- B — RQ2 How soon can filling be flagged, and how many top-ranked alerts are real?
- A+B — RQ3 Do several seasons separate fallow from abandonment, and does radar hold up when clear images are few?
- C — RQ4 Which alert ranking gives the most confirmed cases per field visit?

A land-state model **B** change detection, validation
C pipeline, dashboard, ranking

SCOPE

CORE Gampaha, Kalutara; Maha 2022/23 to 2026/27; four-state map, alerts, dashboard

STRETCH NISAR L-band test, dry-zone transfer, subsidy view

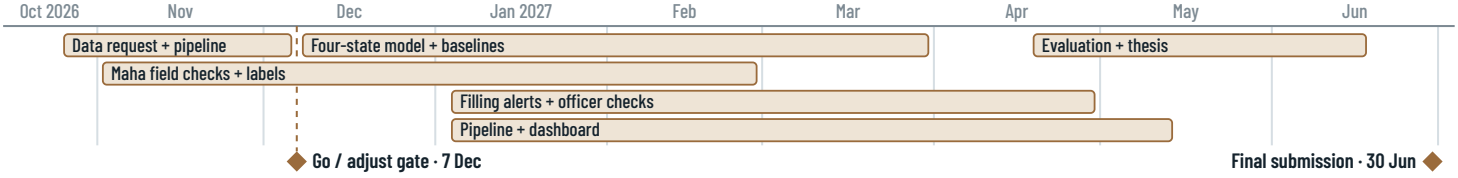
OUT Whole island, legal rulings, drones, yield estimates

MAIN RISKS

- ▲ **Department lists may not come.** Field labels; one Service Centre checks 50 alerts.
- ▲ **Idle and other crop may look alike.** Merge into "not paddy" if F1 < 0.6.
- ▲ **Legal and illegal filling look the same.** Alerts only; officers check permits.

PLAN

proposed; dates assumed, adjust to faculty deadlines



SOURCES Agrarian Development Act 2000 · News 1st 2025 · Ada Derana 2026 · Auditor General 2018 · Agrarian Dept report 2023 · Özdoğan et al. 2025 · Sankavi et al. 2025 · Wanninayaka et al. 2020 · Jayasooriya et al. 2023 · Kasture et al. 2025 · Copernicus 2026 · Census and Statistics

Sheet 3
7.09°N 80.00°E · Gampaha